
TR82xx ESC Register Description

Document Version r1.00

Release Date 2024-11-17



Preface

Overview

This document provides a detailed description of the registers for the TR82xx series EtherCAT slave chips.

Target Audience

This guide is primarily intended for the following engineers:

- Chip Maintenance Support Engineers
- Product Hardware Development/Testing/Maintenance Engineers

Symbol Conventions

The following symbols may appear in this document, and their meanings are as follows:

Symbol	Description
 危险	Used to warn of imminent danger situations that, if not avoided, will result in death or serious personal injury.
 警告	Used to warn of potential danger situations that, if not avoided, could result in death or serious personal injury.
 小心	Used to warn of potential danger situations that, if not avoided, may result in moderate or minor personal injury.
 注意	Used to convey safety warning information regarding equipment or the environment, which, if not avoided, may result in equipment damage, data loss, reduced performance, or other unpredictable outcomes. "Note" does not involve personal injury.
 说明	Used to highlight important/key information, best practices, tips, and tricks. "Note" is not a safety warning and does not involve personal injury, equipment, or environmental damage information.

Change Log

The change log accumulates the descriptions of each document update. The latest version of the document includes all updates from previous versions.

Modification Date	Version	Issue Number	Description of Changes	Author of Changes
2024.11.17	r1.00		Initial Version	Guoqing

Table of Contents

1. ESC Register Map	9
1.1 Register Type (0x0000)	9
1.2 Register Revision (0x0001)	9
1.3 Register Build (0x0002:0x0003)	9
1.4 Register FMMUs supported (0x0004)	9
1.5 Register SyncManagers supported (0x0005)	9
1.6 Register RAM Size (0x0006)	9
1.7 Register Port Descriptor (0x0007)	9
1.8 Register ESC Features supported (0x0008:0x0009)	9
1.9 Register Configured Station Address (0x0010:0x0011)	10
1.10 Register Configured Station Alias (0x0012:0x0013)	10
1.11 Register Write Register Enable (0x0020)	10
1.12 Register Write Register Protection (0x0021)	10
1.13 Register ESC Write Enable (0x0030)	10
1.14 Register ESC Write Protection (0x0031)	10
1.15 Register ESC Reset ECAT (0x0040)	10
1.16 Register ESC Reset PDI (0x0041)	11
1.17 Register ESC DL Control (0x0100:0x0103)	11
1.18 Register Physical Read/Write Offset (0x0108:0x0109)	12
1.19 Register ESC DL Status (0x0110:0x0111)	12
1.20 Register AL Control (0x0120:0x0121)	12
1.21 Register AL Status (0x0130:0x0131)	13
1.22 Register AL Status Code (0x0134:0x0135)	13
1.23 Register RUN LED Override (0x0138)	13
1.24 Register ERR LED Override (0x0139)	13
1.25 Register PDI Control (0x0140)	13
1.26 Register ESC Configuration (0x0141)	14
1.27 Register ESC Configuration (0x0142:0x0143)	14
1.28 Register PDI Configuration Register Overview	14
1.29 Register PDI Digital I/O(0x140:0x4) configuration (0x0150)	15
1.30 Register PDI Digital I/O(0x140:0x4) extended configuration (0x0152:0x0153)	16
1.31 Register PDI SPI(0x140:0x5) /SQI(0x140:0x83) Slave Configuration (0x0150)	16

1.32 Register PDI SPI(0x140:0x5)/SQI(0x140:0x83) Slave extended configuration (0x0152)	16
1.33 Register PDI SPI(0x140:0x5) Slave extended configuration (0x0153)	16
1.34 Register PDI SQI(0x140:0x83) Slave extended configuration (0x0153).....	17
1.35 Register PDI Oversampling(0x140:0x6) configuration (0x0150)	17
1.36 Register PDI Oversampling(0x140:0x6) extended configuration (0x0152:0x153)	17
1.37 Register PDI asynchronous(0x140:0x8/0x9) Microcontroller Configuration (0x0150)	17
1.38 Register PDI Asynchronous(0x140:0x8/0x9) Microcontroller extended Configuration (0x0152:0x0153)	17
1.39 Register PDI HBI (0x140:0x90:0x97) Microcontroller Configuration (0x0150).....	17
1.40 Register PDI HBI (0x140:0x90:0x97) Microcontroller extended configuration (0x0152:0x0153) .	18
1.41 Register Sync/Latch[1:0] PDI Configuration (0x0151)	18
1.42 Register Digital Output WD expiration Configuration (0x01F0:0x1F7)	20
1.43 Register Digital Input Filter Configuration (0x01F8).....	20
1.44 Register PDI AL Event Mask (0x0204:0x0207).....	20
1.45 Register ECAT Event Request (0x0210:0x0211)	20
1.46 Register AL Event Request (0x0220:0x0223).....	20
1.47 Register RX Error Counter Port y (0x0300+y*2:0x0301+y*2)	21
1.48 Register Forwarded RX Error Counter Port y (0x0308+y).....	21
1.49 Register ECAT Processing Unit Error Counter (0x030C)	21
1.50 Register PDI Error Counter (0x030D).....	21
1.51 Register Microcontroller PDI Error Code (0x030E).....	21
1.52 Register Lost Link Counter Port y (0x0310+y)	21
1.53 Register RX Frame Counter Port y (0x0320+y*4:0x321+y*4)	22
1.54 Register TX Frame Counter Port y (0x0320+y*4:0x321+y*4).....	22
1.55 Register Watchdog Divider (0x0400:0x0401).....	22
1.56 Register Watchdog Time PDI (0x0410:0x0411)	22
1.57 Register Watchdog Time Process Data (0x0420:0x0421).....	22
1.58 Register Watchdog Status Process Data (0x0440:0x0441).....	22
1.59 Register Watchdog Counter Process Data (0x0442)	22
1.60 Register Watchdog Counter PDI (0x0443).....	22
1.61 Register EEPROM Configuration (0x0500).....	22
1.62 Register EEPROM PDI Access State (0x0501)	22
1.63 Register EEPROM Control/Status (0x0502:0x0503)	23
1.64 Register EEPROM Address (0x0504:0x0507).....	23

1.65 Register EEPROM Data (0x0508:0x050F [0x0508:0x050B])	23
1.66 Register MII Management Control/Status (0x0510:0x0511)	23
1.67 Register MII Management Control/Status (0x0510:0x0511)	24
1.68 Register PHY Address (0x0512)	25
1.69 Register PHY Register Address (0x0513)	25
1.70 Register PHY Data (0x0514:0x0515)	25
1.71 Register MII Management ECAT Access State (0x0516).....	25
1.72 Register MII Management PDI Access State (0x0517)	25
1.73 Register Logical Start address FMMU y (0x06y0:0x06y3)	25
1.74 Register Length FMMU y (0x06y4:0x06y5)	25
1.75 Register Start bit FMMU y in logical address space (0x06y6)	25
1.76 Register Stop bit FMMU y in logical address space (0x06y7)	25
1.77 Register Physical Start address FMMU y (0x06y8-0x06y9)	25
1.78 Register Physical Start bit FMMU y (0x06yA)	26
1.79 Register Type FMMU y (0x06yB)	26
1.80 Register Activate FMMU y (0x06yC).....	26
1.81 Register Reserved FMMU y (0x06yD:0x06yF)	26
1.82 Register physical Start Address SyncManager y (0x0800+y*8:0x0801+y*8)	26
1.83 Register Length SyncManager y (0x0802+y*8:0x0803+y*8).....	26
1.84 Register Control Register SyncManager y (0x0804+y*8)	26
1.85 Register Status Register SyncManager y (0x0805+y*8)	26
1.86 Register Activate SyncManager y (0x0806+y*8)	27
1.87 Register PDI Control SyncManager y (0x0807+y*8)	27
1.88 Register Receive Time Port 0 (0x0900:0x0903).....	27
1.89 Register Receive Time Port 1 (0x0904:0x0907).....	27
1.90 Register Receive Time Port 2 (0x0908:0x090B).....	28
1.91 Register Receive Time Port 3 (0x090C:0x090F).....	28
1.92 Register Receive Time ECAT Processing Unit (0x0918:0x091F).....	28
1.93 Register System Time (0x0910:0x0913 [0x0910:0x0917])	29
1.94 Register System Time Offset (0x0920:0x0927)	29
1.95 Register System Time Delay (0x0928:0x092B)	29
1.96 Register System Time Difference (0x092C:0x092F).....	29
1.97 Register Speed Counter Start (0x0930:0x931)	29
1.98 Register Speed Counter Diff (0x0932:0x933)	29

1.99 Register System Time Difference Filter Depth (0x0934)	29
1.100 Register Speed Counter Filter Depth (0x0935).....	30
1.101 Register Receive Time Latch Mode (0x0936)	30
1.102 Register Cyclic Unit Control (0x0980)	30
1.103 Register Activation register (0x0981)	30
1.104 Register Pulse Length of SyncSignals (0x0982:0x983).....	31
1.105 Register Pulse Length of SyncSignals (0x0982:0x983).....	31
1.106 Register Activation Status (0x0984)	31
1.107 Register SYNC0 Status (0x098E)	31
1.108 Register SYNC1 Status (0x098F).....	31
1.109 Register Start Time Cyclic Operation (0x0990:0x0993 [0x0990:0x0997])	31
1.110 Register Next SYNC1 Pulse (0x0998:0x099B [0x0998:0x099F])	31
1.111 Register SYNC0 Cycle Time (0x09A0:0x09A3).....	31
1.112 Register SYNC1 Cycle Time (0x09A4:0x09A7).....	31
1.113 Register Latch0 Control (0x09A8).....	32
1.114 Register Latch1 Control (0x09A9).....	32
1.115 Register Latch0 Status (0x09AE).....	32
1.116 Register Latch1 Status (0x09AF)	32
1.117 Register Latch0 Time Positive Edge (0x09B0:0x09B7).....	33
1.118 Register Latch0 Time Negative Edge (0x09B8:0x09BF)	33
1.119 Register Latch1 Time Positive Edge (0x09C0:0x09C7).....	33
1.120 Register Latch1 Time Negative Edge (0x09C8:0x09CF)	33
1.121 Register EtherCAT Buffer Change Event Time (0x09F0:0x09F3)	33
1.122 Register PDI Buffer Start Event Time (0x09F8:0x09FB)	33
1.123 Register PDI Buffer Change Event Time (0x09FC:0x09FF)	33
1.124 Register Power-On Values TR8211K (0x0E00:0x0E01)	33
1.125 Register Power-On Values TR8253/TR8253L (0x0E00:0x0E01).....	34
1.126 Register Vendor ID (0x0E08:0x0E0F)	34
1.127 Register Digital I/O Output Data (0x0F00:0x0F03)	34
1.128 Register General Purpose Outputs (0x0F10:0x0F17)	34
1.129 Register General Purpose Inputs (0x0F18:0x0F1F)	34
1.130 User RAM (0x0F80:0x0FFF).....	35
1.131 Digital I/O Input Data (0x1000:0x1003).....	35
1.132 Process Data RAM (0x1000:0xFFFF)	35

2.ESC ConfigData36

1. ESC Register Map

1.1 Register Type (0x0000)

Bit	Description	ECAT	PDI	Reset Value
7:0	Type of EtherCAT controller	r/-	r/-	0xA6

1.2 Register Revision (0x0001)

Bit	Description	ECAT	PDI	Reset Value
7:0	Revision of EtherCAT controller. IP Core: [7:4] = minor version Y, [3:0] = maintenance version X	r/-	r/-	0

1.3 Register Build (0x0002:0x0003)

Bit	Description	ECAT	PDI	Reset Value
7:0	Actual build of EtherCAT controller. IP Core: [7:4] = minor version Y, [3:0] = maintenance version Z	r/-	r/-	0

1.4 Register FMMUs supported (0x0004)

Bit	Description	ECAT	PDI	Reset Value
7:0	Number of supported FMMU channels (or entities) of the EtherCAT Slave Controller.	r/-	r/-	8

1.5 Register SyncManagers supported (0x0005)

Bit	Description	ECAT	PDI	Reset Value
7:0	Number of supported SyncManager channels (or entities) of the EtherCAT Slave Controller	r/-	r/-	8

1.6 Register RAM Size (0x0006)

Bit	Description	ECAT	PDI	Reset Value
7:0	Process Data RAM size supported by the EtherCAT Slave Controller in KByte	r/-	r/-	8

1.7 Register Port Descriptor (0x0007)

Bit	Description	ECAT	PDI	Reset Value
	Port configuration: 00: Not implemented 01: Not configured (SII EEPROM) 10: EBUS 11: MII / RMII / RGMII			
1:0	Port 0	r/-	r/-	ESC and ESC configuration dep.
3:2	Port 1	r/-	r/-	
5:4	Port 2	r/-	r/-	
7:6	Port 3	r/-	r/-	

1.8 Register ESC Features supported (0x0008:0x0009)

Bit	Description	ECAT	PDI	Reset Value
0	FMMU Operation: 0: Bit oriented 1: Byte oriented	r/-	r/-	0
1	Reserved	r/-	r/-	0
2	Distributed Clocks: 0: Not available 1: Available	r/-	r/-	1
3	Distributed Clocks (width): 0: 32 bit 1: 64 bit	r/-	r/-	1
4	Low Jitter EBUS: 0: Not available, standard jitter 1: Available, jitter minimized	r/-	r/-	0
5	Enhanced Link Detection EBUS: 0: Not available 1: Available	r/-	r/-	0
6	Enhanced Link Detection MII: 0: Not available 1: Available	r/-	r/-	1
7	Separate Handling of FCS Errors: 0: Not supported 1: Supported, frames with wrong FCS and additional nibble will be counted separately in Forwarded RX Error Counter	r/-	r/-	1

8	Enhanced DC SYNC Activation 0: Not available 1: Available NOTE: This feature refers to registers 0x981[7:3], 0x0984	r/-	r/-	1
9	EtherCAT LRW command support: 0: Supported 1: Not supported	r/-	r/-	0
10	EtherCAT read/write command support (BRW, APRW, FPRW): 0: Supported 1: Not supported	r/-	r/-	0
11	Fixed FMMU/SyncManager configuration 0: Variable configuration 1: Fixed configuration (refer to documentation of supporting ESCs)	r/-	r/-	0
15:12	Reserved	r/-	r/-	0

1.9 Register Configured Station Address (0x0010:0x0011)

Bit	Description	ECAT	PDI	Reset Value
15:0	Address used for node addressing (FPxx commands)	r/w	r/-	0

1.10 Register Configured Station Alias (0x0012:0x0013)

Bit	Description	ECAT	PDI	Reset Value
15:0	Alias Address used for node addressing (FPxx commands). The use of this alias is activated by Register DL Control Bit 24 (0x0100.24/0x0103.0) NOTE: EEPROM value is only taken over at first EEPROM load after power-on or reset.	r/-	r/w	0 until first EEPROM load, then EEPROM ADR 0x0004

1.11 Register Write Register Enable (0x0020)

Bit	Description	ECAT	PDI	Reset Value
0	If write register protection is enabled, this register has to be written in the same Ethernet frame (value does not care) before other writes to this station are allowed. Write protection is still active after this frame (if Write Register Protection register is not changed).	r/w	r/-	0
7:1	Reserved, write 0	r/-	r/-	0

1.12 Register Write Register Protection (0x0021)

Bit	Description	ECAT	PDI	Reset Value
0	Write register protection: 0: Protection disabled 1: Protection enabled Registers 0x0000-0x0137, 0x013A-0x0F0F are write protected, except for 0x0030.	r/w	r/-	0
7:1	Reserved, write 0	r/-	r/-	0

1.13 Register ESC Write Enable (0x0030)

Bit	Description	ECAT	PDI	Reset Value
0	If ESC write protection is enabled, this register has to be written in the same Ethernet frame (value does not care) before other writes to this station are allowed. ESC write protection is still active after this frame (if ESC Write Protection register is not changed).	r/w	r/-	0
7:1	Reserved, write 0	r/-	r/-	0

1.14 Register ESC Write Protection (0x0031)

Bit	Description	ECAT	PDI	Reset Value
0	Write protect: 0: Protection disabled 1: Protection enabled All areas are write protected, except for 0x0030.	r/w	r/-	0
7:1	Reserved, write 0	r/-	r/-	0

1.15 Register ESC Reset ECAT (0x0040)

Bit	Description	ECAT	PDI	Reset Value
Write				
7:0	A reset is asserted after writing 0x52 ('R '), 0x45 ('E') and 0x53 ('S') in this register with 3 consecutive frames.	r/w	r/-	0
Read				
1:0	Progress of the reset procedure: 01: after writing 0x52 10: after writing 0x45 (if 0x52 was written before)	r/w	r/-	0

00:	else			
7:2	Reserved, write 0	r/-	r/-	0

1.16 Register ESC Reset PDI (0x0041)

Bit	Description	ECAT	PDI	Reset Value
Write				
7:0	A reset is asserted after writing 0x52 ('R'), 0x45 ('E') and 0x53 ('S') in this register with 3 consecutive commands.	r/-	r/w	0
Read				
1:0	Progress of the reset procedure: 01: after writing 0x52 10: after writing 0x45 (if 0x52 was written before) 00: else	r/-	r/w	0
7:2	Reserved, write 0	r/-	r/-	0

1.17 Register ESC DL Control (0x0100:0x0103)

Bit	Description	ECAT	PDI	Reset Value
0	Forwarding rule: 0: EtherCAT frames are processed, Non EtherCAT frames are forwarded without processing 1: EtherCAT frames are processed, Non-EtherCAT frames are destroyed The source MAC address is changed for every frame (SOURCE_MAC[1] is set to 1 – locally administered address) regardless of the forwarding rule.	r/w	r/-	1
1	Temporary use of settings in Register 0x101: 0: permanent use 1: use for about 1 second, then revert to previous settings	r/w	r/-	0
7:2	Reserved, write 0	r/-	r/-	0
9:8	Loop Port 0: 00: Auto 01: Auto Close 10: Open 11: Closed NOTE: Loop open means sending/receiving over this port is enabled, loop closed means sending/receiving is disabled and frames are forwarded to the next open port internally. Auto: loop closed at link down, opened at link up Auto Close: loop closed at link down, opened with writing 01 again after link up (or receiving a valid Ethernet frame at the closed port) Open: loop open regardless of link state Closed: loop closed regardless of link state	r/w*	r/-	0
11:10	Loop Port 1: 00: Auto 01: Auto Close 10: Open 11: Closed	r/w*	r/-	00
13:12	Loop Port 2: 00: Auto 01: Auto Close 10: Open 11: Closed	r/w*	r/-	00
15:14	Loop Port 3: 00: Auto 01: Auto Close 10: Open 11: Closed	r/w*	r/-	00
18:16	RX FIFO Size (ESC delays start of forwarding until FIFO is at least half full). RX FIFO Size/RX delay reduction** : Value: EBUS: MII: 0: -50 ns -40 ns 1: -40 ns -40 ns 2: -30 ns -40 ns 3: -20 ns -40 ns 4: -10 ns no change 5: no change no change 6: no change no change 7: default default NOTE: EEPROM value is only taken over at first EEPROM load after power-on or reset	r/w	r/-	7
23:19	Reserved, write 0	r/w	r/-	0
24	Station alias: 0: Ignore Station Alias	r/w	r/-	0

	1: Alias can be used for all configured address command types (FPRD, FPWR,...)			
31:25	Reserved, write 0	r/-	r/-	0

* Loop configuration changes are delayed until the end of a currently received or transmitted frame at the port.

** The possibility of RX FIFO Size reduction depends on the clock source accuracy of the ESC and of every connected EtherCAT/Ethernet devices (master, slave, etc.). RX FIFO Size of 7 is sufficient for 100ppm accuracy, FIFO Size 0 is possible with 25ppm accuracy (frame size of 1518/1522 Byte).

1.18 Register Physical Read/Write Offset (0x0108:0x0109)

Bit	Description	ECAT	PDI	Reset Value
15:0	Offset of R/W Commands (FPRW, APRW) between Read address and Write address. RD_ADR = ADR and WR_ADR = ADR + R/W-Offset	r/w	r/-	0

1.19 Register ESC DL Status (0x0110:0x0111)

Bit	Description	ECAT	PDI	Reset Value
0	PDI operational/EEPROM loaded correctly: 0: EEPROM not loaded, PDI not operational (no access to Process Data RAM) 1: EEPROM loaded correctly, PDI operational (access to Process Data RAM)	r*/-	r/-	0
1	PDI Watchdog Status: 0: Watchdog expired 1: Watchdog reloaded	r*/-	r/-	0
2	Enhanced Link detection: 0: Deactivated for all ports 1: Activated for at least one port NOTE: EEPROM value is only taken over at first EEPROM load after power-on or reset	r*/-	r/-	1 until first EEPROM load, then EEPROM ADR 0x0000.9 or 0x0000[15:12]
3	Reserved	r*/-	r/-	0
4	Physical link on Port 0: 0: No link 1: Link detected	r*/-	r/-	0
5	Physical link on Port 1: 0: No link 1: Link detected	r*/-	r/-	0
6	Physical link on Port 2: 0: No link 1: Link detected	r*/-	r/-	0
7	Physical link on Port 3: 0: No link 1: Link detected	r*/-	r/-	0
8	Loop Port 0: 0: Open 1: Closed	r*/-	r/-	0
9	Communication on Port 0: 0: No stable communication 1: Communication established	r*/-	r/-	0
10	Loop Port 1: 0: Open 1: Closed	r*/-	r/-	0
11	Communication on Port 1: 0: No stable communication 1: Communication established	r*/-	r/-	0
12	Loop Port 2: 0: Open 1: Closed	r*/-	r/-	0
13	Communication on Port 2: 0: No stable communication 1: Communication established	r*/-	r/-	0
14	Loop Port 3: 0: Open 1: Closed	r*/-	r/-	0
15	Communication on Port 3: 0: No stable communication 1: Communication established	r*/-	r/-	0

* Reading DL Status register from ECAT clears ECAT Event Request 0x0210[2].

1.20 Register AL Control (0x0120:0x0121)

Bit	Description	ECAT	PDI	Reset Value
3:0	Initiate State Transition of the Device State Machine: 1: Request Init State 3: Request Bootstrap State 2: Request Pre-Operational State 4: Request Safe-Operational State 8: Request Operational State	r/(w)	r	1

4	Error Ind Ack: 0: No Ack of Error Ind in AL status register 1: Ack of Error Ind in AL status register	r/(w)	r	0
5	Device Identification: 0: No request 1: Device Identification request	r/(w)	r	0
15:6	Reserved, write 0	r/(w)	r	0

NOTE: AL Control register behaves like a mailbox if Device Emulation is off (0x0140.8=0): The PDI has to read the AL Control register after ECAT has written it. Otherwise ECAT cannot write again to the AL Control register. After Reset, AL Control register can be written by ECAT. (Regarding mailbox functionality, both registers 0x0120 and 0x0121 are equivalent, e.g. reading 0x0121 is sufficient to make this register writeable again.) If Device Emulation is on, the AL Control register can always be written, its content is copied to the AL Status register.

1.21 Register AL Status (0x0130:0x0131)

Bit	Description	ECAT	PDI	Reset Value
3:0	Actual State of the Device State Machine: 1: Init State 3: Request Bootstrap State 2: Pre-Operational State 4: Safe-Operational State 8: Operational State	r*/-	r/(w)	1
4	Error Ind: 0: Device is in State as requested or Flag cleared by command 1: Device has not entered requested State or changed State as result of a local action	r*/-	r/(w)	0
5	Device Identification: 0: Device Identification not valid 1: Device Identification loaded	r*/-	r/(w)	0
15:6	Reserved, write 0	r*/-	r/(w)	0

NOTE: AL Status register is only writable from PDI if Device Emulation is off (0x0140.8=0), otherwise AL Status register will reflect AL Control register values.

* Reading AL Status from ECAT clears ECAT Event Request 0x0210[3].

1.22 Register AL Status Code (0x0134:0x0135)

Bit	Description	ECAT	PDI	Reset Value
15:0	AL Status Code	r/-	r/w	0

1.23 Register RUN LED Override (0x0138)

Bit	Description	ECAT	PDI	Reset Value
3:0	LED code: (FSM State) 0x0: Off (1-Init) 0x1-0xC: Flash 1x – 12x (4-SafeOp1x) 0xD: Blinking (2-PreOp) 0xE: Flickering (3-Bootstrap) 0xF: On (8-Op)	r/w	r/w	0
4	Enable Override: 0: Override disabled 1: Override enabled	r/w	r/w	0
7:5	Reserved, write 0	r/w	r/w	0

NOTE: Changes to AL Status register (0x0130) with valid values will disable RUN LED Override (0x0138[4]=0). The value read in this register always reflects current LED output.

1.24 Register ERR LED Override (0x0139)

Bit	Description	ECAT	PDI	Reset Value
3:0	LED code: 0x0: Off 0x1-0xC: Flash 1x – 12x 0xD: Blinking 0xE: Flickering 0xF: On	r/w	r/w	0
4	Enable Override: 0: Override disabled 1: Override enabled	r/w	r/w	0
7:5	Reserved, write 0	r/w	r/w	0

NOTE: New error conditions will disable ERR LED Override (0x0139[4]=0). The value read in this register always reflects current LED output.

1.25 Register PDI Control (0x0140)

Bit	Description	ECAT	PDI	Reset Value
-----	-------------	------	-----	-------------

7:0	Process data interface: 0x00: Interface deactivated (no PDI) 0x04: Digital I/O 0x05: Beckhoff SPI Slave 0x08: 16 Bit asynchronous Microcontroller interface 0x09: 8 Bit asynchronous Microcontroller interface 0x83: Triductor SQI Slave 0x90: HBI Multiplexed 1 Phase 8-bit 0x91: HBI Multiplexed 1 Phase 16-bit 0x92: HBI Multiplexed 2 Phase 8-bit 0x93: HBI Multiplexed 2 Phase 16-bit 0x94: HBI Indexed 8-bit 0x95: HBI Indexed 16-bit 0x96: HBI Demultiplexed 8-bit 0x97: HBI Demultiplexed 16-bit Others: Reserved	r/-	r/-	0, later EEPROM ADR 0x0000
-----	--	-----	-----	-------------------------------

1.26 Register ESC Configuration (0x0141)

Bit	Description	ECAT	PDI	Reset Value
0	Device emulation (control of AL status): 0: AL status register has to be set by PDI 1: AL status register will be set to value written to AL control register	r/-	r/-	0, later EEPROM ADR 0x0000
1	Enhanced Link detection all ports: 0: disabled (if bits [7:4]=0) 1: enabled at all ports (overrides bits [7:4])	r/-	r/-	1, later EEPROM ADR 0x0000
2	Distributed Clocks SYNC Out Unit: 0: disabled (power saving) 1: enabled	r/-	r/-	0, later EEPROM ADR 0x0000
3	Distributed Clocks Latch In Unit: 0: disabled (power saving) 1: enabled	r/-	r/-	0, later EEPROM ADR 0x0000
4	Enhanced Link port 0: 0: disabled (if bit 1=0) 1: enabled	r/-	r/-	1, later EEPROM ADR 0x0000
5	Enhanced Link port 1: 0: disabled (if bit 1=0) 1: enabled	r/-	r/-	1, later EEPROM ADR 0x0000
6	Enhanced Link port 2: 0: disabled (if bit 1=0) 1: enabled	r/-	r/-	1, later EEPROM ADR 0x0000
7	Enhanced Link port 3: 0: disabled (if bit 1=0) 1: enabled	r/-	r/-	1, later EEPROM ADR 0x0000

1.27 Register ESC Configuration (0x0142:0x143)

Bit	Description	ECAT	PDI	Reset Value
4:0	Digital input filter time: 0x00: off 0x01: 100us 0x02: 200us 0x03: 500us 0x04-0x1F: 1ms~28ms(one step plus 1ms)	r/-	r/-	0, later EEPROM ADR 0x0005
5	Reserved, set EEPROM value 0	r/-	r/-	0, later EEPROM ADR 0x0005
6	STATE_RUNLED Mode Select: 0: disabled 1: enabled	r/-	r/-	0, later EEPROM ADR 0x0005
7	Reserved, set EEPROM value 0	r/-	r/-	0, later EEPROM ADR 0x0005
9:8	When Watchdog expiration, Digital Output: 00: High Resistance 01: Keep last data 10: Zero 11: One	r/-	r/-	0, later EEPROM ADR 0x0005
13:10	Reserved, set EEPROM value 0	r/-	r/-	0, later EEPROM ADR 0x0005
14	ERRLED Enable: 0: disabled 1: enabled	r/-	r/-	1, later EEPROM ADR 0x0000
15	Reserved, set EEPROM value 0	r/-	r/-	0, later EEPROM ADR 0x0005

1.28 Register PDI Configuration Register Overview

PDI number	PDI name	Configuration registers	CHIP
0x04	Digital I/O	0x150,0x152:01x153,0x1f0:0x1f8	TR8211/TR8211K
0x05	SPI Slave	0x150,0x152:01x153	TR8253/TR8253L

0x06	Oversampling I/O	0x150,0x152:01x153	
0x07	EtherCAT Bridge (port 3)	0x150,0x152:01x153	
0x08/0x09	8/16 Bit asynchronous Microcontroller interface	0x150,0x152:01x153	TR8211/TR8211K
0x83	SQI Slave	0x150,0x152:01x153	
0x90/0x91	HBI Multiplexed 1 Phase 8/16-bit	0x150,0x152:01x153	TR8211/TR8211K
0x92/0x93	HBI Multiplexed 2 Phase 8/16-bit	0x150,0x152:01x153	TR8253/TR8253L
0x94/0x95	HBI Indexed 8/16-bit	0x150,0x152:01x153	
0x96/0x97	HBI Demultiplexed 8/16-bit	0x150,0x152:01x153	TR8211/TR8211K

1.29 Register PDI Digital I/O(0x140:0x4) configuration (0x0150)

Bit	Description	ECAT	PDI	Reset Value
0	OUTVALID polarity: 0: Active high 1: Active low	r/-	r/-	0, later EEPROM ADR 0x0001
1	OUTVALID mode: 0: Output event signaling 1: Process Data Watchdog trigger (WD_TRIG) signaling on OUTVALID pin (see SyncManager) Output data is updated if watchdog is triggered. Overrides 0x0150[7:6]	r/-	r/-	
2	Unidirectional/Bidirectional mode: 0: Unidirectional mode: input/output direction of pins configured individually 1: Bidirectional mode: all I/O pins are bidirectional, direction configuration is ignored	r/-	r/-	
3	Watchdog behavior: 0: Outputs are reset immediately after watchdog expires 1: Outputs are reset with next output event that follows watchdog expiration	r/-	r/-	
5:4	Input DATA is sampled at 00: Start of Frame2 01: Rising edge of LATCH_IN 10: DC SYNC0 event ² 11: DC SYNC1 event ²	r/-	r/-	00, later EEPROM ADR 0x0001
7:6	Output DATA is updated at 00: End of Frame 01: reserved 10: DC SYNC0 event 11: DC SYNC1 event If 0x0150[1]=1, output DATA is updated at Process Data Watchdog trigger event (0x0150[7:6] are ignored)	r/-	r/-	

1.30 Register PDI Digital I/O(0x140:0x4) extended configuration (0x0152:0x0153)

Bit	Description	ECAT	PDI	Reset Value
	Digital I/Os are configured in pairs as inputs or outputs: 0: Input 1: Output NOTE: Reserved in bidirectional mode, set to 0. Configuration bits for unavailable I/Os are reserved, set to 0.			
0	Direction of I/O[1:0]	r/-	r/-	0, later EEPROM ADR 0x0003
1	Direction of I/O[3:2]	r/-	r/-	
2	Direction of I/O[5:4]	r/-	r/-	
3	Direction of I/O[7:6]	r/-	r/-	
4	Direction of I/O[9:8]	r/-	r/-	
5	Direction of I/O[11:10]	r/-	r/-	
6	Direction of I/O[13:12]	r/-	r/-	
7	Direction of I/O[15:14]	r/-	r/-	
8	Direction of I/O[17:16]	r/-	r/-	
9	Direction of I/O[19:18]	r/-	r/-	
10	Direction of I/O[21:20]	r/-	r/-	
11	Direction of I/O[23:22]	r/-	r/-	
12	Direction of I/O[25:24]	r/-	r/-	
13	Direction of I/O[27:26]	r/-	r/-	
14	Direction of I/O[29:28]	r/-	r/-	
15	Direction of I/O[31:30]	r/-	r/-	

1.31 Register PDI SPI(0x140:0x5) /SQI(0x140:0x83) Slave Configuration (0x0150)

Bit	Description	ECAT	PDI	Reset Value
1:0	SPI mode: 00: SPI mode 0 01: SPI mode 1 10: SPI mode 2 11: SPI mode 3 NOTE: SPI mode 3 is recommended for Slave Sample Code NOTE: SPI status flag is not available in SPI modes 0 and 2 with normal data out sample. NOTE: SQI Not Available	r/-	r/-	0, later EEPROM ADR 0x0001
3:2	SPI_IRQ/SQI_IRQ output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)	r/-	r/-	
4	Reserved, set EEPROM value 0	r/-	r/-	
5	Data Out sample mode: 0: Normal sample (SPI_DO and SPI_DI are sampled at the same SPI_CLK edge) 1: Late sample (SPI_DO and SPI_DI are sampled at different SPI_CLK edges) NOTE: SQI Not Available	r/-	r/-	
7:6	Reserved, set EEPROM value 0	r/-	r/-	

1.32 Register PDI SPI(0x140:0x5)/SQI(0x140:0x83) Slave extended configuration (0x0152)

Bit	Description	ECAT	PDI	Reset Value
	General Purpose I/O Direction 0: Input 1: Output NOTE: TR8253/TR8253L Available			0, later EEPROM ADR 0x0003
0	I/O[1:0]	r/-	r/-	
1	I/O[3:2]	r/-	r/-	
2	I/O[5:4]	r/-	r/-	
3	I/O[7:6]	r/-	r/-	
4	I/O[9:8]	r/-	r/-	
5	I/O[11:10]	r/-	r/-	
6	I/O[13:12]	r/-	r/-	
7	I/O[15:14]	r/-	r/-	

1.33 Register PDI SPI(0x140:0x5) Slave extended configuration (0x0153)

Bit	Description	ECAT	PDI	Reset Value
	General Purpose I/O Buffer Type 0: Push Pull 1: Open Drain NOTE: TR8253/TR8253L Available			
0	I/O[1:0]	r/-	r/-	

1	I/O[3:2]	r/-	r/-	
2	I/O[5:4]	r/-	r/-	
3	I/O[7:6]	r/-	r/-	
4	I/O[9:8]	r/-	r/-	
5	I/O[11:10]	r/-	r/-	
6	I/O[13:12]	r/-	r/-	
7	I/O[15:14]	r/-	r/-	

1.34 Register PDI SQI(0x140:0x83) Slave extended configuration (0x0153)

Bit	Description	ECAT	PDI	Reset Value
3:0	Length of byte(s) from Address to the first Write access Length = Dummy bytes + 2(Address bytes) – 1 Min Value :1	r/-	r/-	0, later EEPROM ADR 0x0003
7:4	Length of byte(s) from Address to the first Read access Length = Dummy bytes + 2(Address bytes) + 1(Length bytes) – 1 Min Value :4	r/-	r/-	

1.35 Register PDI Oversampling(0x140:0x6) configuration (0x0150)

Bit	Description	ECAT	PDI	Reset Value
0	Oversampling Output Enable Pin Polarity 0: Active Low 1: Active High	r/-	r/-	00, later EEPROM ADR 0x0001
1	Second Channel Enable: 0: Disable 1: Enable	r/-	r/-	
3:2	Oversampling Direction: 01: Input 10: Output Others: Reserved	r/-	r/-	
7:4	Reserved, set EEPROM value 0	r/-	r/-	

1.36 Register PDI Oversampling(0x140:0x6) extended configuration (0x0152:0x153)

Bit	Description	ECAT	PDI	Reset Value
7:0	When CycleCounter Error or watchdog run out and 0xF00 = 0x6 The Value of Channel 0 output	r/-	r/-	00, later EEPROM ADR 0x0003
15:8	When CycleCounter Error or watchdog run out and 0xF01 = 0x6 The Value of Channel 1 output	r/-	r/-	

1.37 Register PDI asynchronous(0x140:0x8/0x9) Microcontroller Configuration (0x0150)

Bit	Description	ECAT	PDI	Reset Value
1:0	BUSY output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high) NOTE: Push-Pull: no CS → not BUSY (driven) Open Drain/Source: no CS → BUSY open	r/-	r/-	00, later EEPROM ADR 0x0001
3:2	IRQ output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)	r/-	r/-	
4	BHE/Byte Enable polarity: 0: Active low 1: Active high	r/-	r/-	
5	Reserved, set EEPROM value 0	r/-	r/-	0, later EEPROM ADR 0x0001
6	Reserved, set EEPROM value 0	r/-	r/-	
7	RD Polarity: 0: Active low 1: Active high	r/-	r/-	

1.38 Register PDI Asynchronous(0x140:0x8/0x9) Microcontroller extended Configuration (0x0152:0x0153)

Bit	Description	ECAT	PDI	Reset Value
0	Read BUSY delay: 0: Normal read BUSY output 1: Delayed read BUSY output	r/-	r/-	0, later EEPROM ADR 0x0003
1	Perform internal write at: 0: End of write access 1: Beginning of write access	r/-	r/-	
15:2	Reserved, set EEPROM value 0	r/-	r/-	

1.39 Register PDI HBI (0x140:0x90:0x97) Microcontroller Configuration (0x0150)

Bit	Description	ECAT	PDI	Reset Value
-----	-------------	------	-----	-------------

0	HBI WAIT_ACK Buffer Type 0: Push-Pull 1: Open Drain	r/-	r/-	0, later EEPROM ADR 0x0001
1	HBI WAIT_ACK Polarity 0: Active low 1: Active high	r/-	r/-	
2	HBI ALE Polarity Configures the polarity of the HBI interface ALEHI and ALELO signals. 0: Active Low Strobe (Address saved on rising edge) 1: Active High Strobe (Address saved on falling edge) NOTE: Only Multiplexed Available	r/-	r/-	
3	HBI Write, Enable Polarity Configures the polarity of the HBI interface write signal. 0: Active Low Write 1: Active High Write Configures the polarity of the HBI interface read/write signal. 0: Active Low Enable 1: Active High Enable	r/-	r/-	
4	HBI Read, Read/Write Polarity Configures the polarity of the HBI interface read signal. 0: Active Low Read 1: Active High Read Configures the polarity of the HBI interface read/write signal. 0: Read when 1, write when 0 (R/nW) 1: Write when 1, read when 0 (W/nR)	r/-	r/-	
5	HBI Chip Select Polarity Configures the polarity of the HBI interface chip select signal. 0: Active Low 1: Active High	r/-	r/-	
6	HBI Read/Write Mode Configures the HBI interface for separate read and write signals or direction and enable signals. 0: Read and Write 1: Direction and Enable	r/-	r/-	
7	HBI ALE Qualification Configures the HBI interface to qualify the ALEHI and ALELO signals with the CS signal. 0: Address input is latched with ALEHI and ALELO 1: Address input is latched with ALEHI and ALELO only when CS is active.	r/-	r/-	

1.40 Register PDI HBI (0x140:0x90:0x97) Microcontroller extended configuration (0x0152:0x0153)

Bit	Description	ECAT	PDI	Reset Value
0	Read WAIT_ACK Removal Delay 0: Normal read WAIT_ACK output 1: Delayed read WAOT_ACK output	r/-	r/-	0, later EEPROM ADR 0x0003
1	Perform Internal Write 0: Following host write cycle (posted) 1: Following host read cycle (non-posted)	r/-	r/-	
2	HBI BE1/BE0 Polarity 0: BE1/BE0 pins active low 1: BE1/BE0 pins active high Note: Multiplexed Mode only	r/-	r/-	
3	IRQ output polarity: 0: active low 1: active high "	r/-	r/-	
4	HBI BE1/BE0 Ignore 0: BE1/BE0 pins activated 1: BE1/BE0 pins ignored,value is 0	r/-	r/-	
15:5	RESERVED	r/-	r/-	

1.41 Register Sync/Latch[1:0] PDI Configuration (0x0151)

Bit	Description	ECAT	PDI	Reset Value
1:0	SYNC0 output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)	r/-	r/-	00, later EEPROM ADR 0x0001
2	SYNC0/LATCH0 configuration*: 0: LATCH0 Input 1: SYNC0 Output	r/-	r/-	
3	SYNC0 mapped to AL Event Request register 0x0220.2: 0: Disabled 1: Enabled	r/-	r/-	0, later EEPROM ADR 0x0001
5:4	SYNC1 output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)	r/-	r/-	00, later EEPROM ADR 0x0001

6	SYNC1/LATCH1 configuration*: 0: LATCH1 input 1: SYNC1 output	r/-	r/-	0, later EEPROM ADR 0x0001
7	SYNC1 mapped to AL Event Request register 0x0220.3: 0: Disabled 1: Enabled	r/-	r/-	0, later EEPROM ADR 0x0001

1.42 Register Digital Output WD expiration Configuration (0x01F0:0x1F7)

Bit	Description	ECAT	PDI	Reset Value
	When Watchdog expiration, Digital Output: 00: High Resistance 01: Keep last data 10: Zero 11: One			
1+2*y:0+2*y	Digital Output Channel y	r/w	r/-	0, later EEPROM ADR 0x0005

1.43 Register Digital Input Filter Configuration (0x01F8)

Bit	Description	ECAT	PDI	Reset Value
15:0	ECAT Event masking of the ECAT Event Request Events for mapping into ECAT event field of EtherCAT frames: 0: Corresponding ECAT Event Request register bit is not mapped 1: Corresponding ECAT Event Request register bit is mapped	r/w	r/-	0

1.44 Register PDI AL Event Mask (0x0204:0x0207)

Bit	Description	ECAT	PDI	Reset Value
31:0	AL Event masking of the AL Event Request register Events for mapping to PDI IRQ signal: 0: Corresponding AL Event Request register bit is not mapped 1: Corresponding AL Event Request register bit is mapped	r/-	r/w	0x00FF:0xFF0F

1.45 Register ECAT Event Request (0x0210:0x0211)

Bit	Description	ECAT	PDI	Reset Value
0	DC Latch event: 0: No change on DC Latch Inputs 1: At least one change on DC Latch Inputs (Bit is cleared by reading DC Latch event times from ECAT for ECAT controlled Latch Units, so that Latch 0/1 Status 0x09AE:0x09AF indicates no event)	r/-	r/-	0
1	Reserved	r/-	r/-	0
2	DL Status event: 0: No change in DL Status 1: DL Status change (Bit is cleared by reading out DL Status 0x0110:0x0111 from ECAT)	r/-	r/-	0
3	AL Status event: 0: No change in AL Status 1: AL Status change (Bit is cleared by reading out AL Status 0x0130:0x0131 from ECAT)	r/-	r/-	0
4	Mirrors values of each SyncManager Status: 0: No Sync Channel 0 event 1: Sync Channel 0 event pending	r/-	r/-	0
5	0: No Sync Channel 1 event 1: Sync Channel 1 event pending			
...	...			
11	0: No Sync Channel 7 event 1: Sync Channel 7 event pending			
15:12	Reserved	r/-	r/-	0

1.46 Register AL Event Request (0x0220:0x0223)

Bit	Description	ECAT	PDI	Reset Value
0	AL Control event: 0: No AL Control Register change 1: AL Control Register has been written ³ (Bit is cleared by reading AL Control register 0x0120:0x0121 from PDI)	r/-	r/-	0
1	DC Latch event: 0: No change on DC Latch Inputs 1: At least one change on DC Latch Inputs (Bit is cleared by reading DC Latch event times from PDI, so that Latch 0/1 Status 0x09AE:0x09AF indicates no event. Available if Latch Unit is PDI controlled)	r/-	r/-	0
2	State of DC SYNC0 (if register 0x0151.3=1): (Bit is cleared by reading SYNC0 status 0x098E from PDI, use only in Acknowledge mode)	r/-	r/-	0
3	State of DC SYNC1 (if register 0x0151.7=1): (Bit is cleared by reading of SYNC1 status 0x098F from PDI, use only in Acknowledge mode)	r/-	r/-	0
4	SyncManager activation register (SyncManager register offset 0x6) changed:	r/-	r/-	0

	0: No change in any SyncManager 1: At least one SyncManager changed (Bit is cleared by reading SyncManager Activation registers 0x0806 etc. from PDI)			
5	EEPROM Emulation: 0: No command pending 1: EEPROM command pending (Bit is cleared by acknowledging the command in EEPROM command register 0x0502 from PDI)	r/-	r/-	0
6	Watchdog Process Data: 0: Has not expired 1: Has expired (Bit is cleared by reading Watchdog Status Process Data 0x0440 from PDI)	r/-	r/-	0
7	Reserved	r/-	r/-	0
8	SyncManager interrupts (SyncManager register offset 0x5, bit [0] or [1]): 0: No SyncManager 0 interrupt 1: SyncManager 0 interrupt pending	r/-	r/-	0
9	0: No SyncManager 1: interrupt			
...	...			
23	0: No SyncManager 15 interrupt 1: SyncManager 15 interrupt pending			
31:24	Reserved	r/-	r/-	0

1.47 Register RX Error Counter Port y (0x0300+y*2:0x0301+y*2)

Bit	Description	ECAT	PDI	Reset Value
7:0	Invalid frame counter of Port y (counting is stopped when 0xFF is reached).	r/ w(clr)	r/-	0
15:8	RX Error counter of Port y (counting is stopped when 0xFF is reached). This is coupled directly to RX ERR of MII interface/EBUS interface.	r/ w(clr)	r/-	0

NOTE: Error Counters 0x0300-0x030B are cleared if one of the RX Error counters 0x0300-0x030B is written. Write value is ignored (write 0).

1.48 Register Forwarded RX Error Counter Port y (0x0308+y)

Bit	Description	ECAT	PDI	Reset Value
7:0	Forwarded error counter of Port y (counting is stopped when 0xFF is reached).	r/ w(clr)	r/-	0

NOTE: Error Counters 0x0300-0x030B are cleared if one of the RX Error counters 0x0300-0x030B is written. Write value is ignored (write 0).

1.49 Register ECAT Processing Unit Error Counter (0x030C)

Bit	Description	ECAT	PDI	Reset Value
7:0	ECAT Processing Unit error counter (counting is stopped when 0xFF is reached). Counts errors of frames passing the Processing Unit (e.g., FCS is wrong or datagram structure is wrong).	r/ w(clr)	r/-	0

NOTE: Error Counter 0x030C is cleared if error counter 0x030C is written. Write value is ignored (write 0).

1.50 Register PDI Error Counter (0x030D)

Bit	Description	ECAT	PDI	Reset Value
7:0	PDI Error counter (counting is stopped when 0xFF is reached). Counts if a PDI access has an interface error.	r/ w(clr)	r/-	0

NOTE: Error Counter 0x030D and Error Code 0x030E are cleared if error counter 0x030D is written. Write value is ignored (write 0).

1.51 Register Microcontroller PDI Error Code (0x030E)

Bit	Description	ECAT	PDI	Reset Value
	µC access which caused last PDI Error.	r/-	r/-	0
0	Busy violation during read access			
1	Busy violation during write access			
2	Addressing error for a read access (A[0]=1 and BHE(act. low)=0)			
3	Addressing error for a write access (A[0]=1 and BHE(act. low)=0)			
7:4	reserved			

NOTE: Error Counter 0x030D and Error Code 0x030E are cleared if error counter 0x030D is written. Write value is ignored (write 0).

1.52 Register Lost Link Counter Port y (0x0310+y)

Bit	Description	ECAT	PDI	Reset Value
7:0	Lost Link counter of Port y (counting is stopped when 0xFF is reached). Counts only if port loop is Auto.	r/ w(clr)	r/-	0

NOTE: Only lost links at open ports are counted. Lost Link Counters 0x0310-0x0313 are cleared if one of the Lost Link Counters 0x0310-0x0313 is written. Write value is ignored (write 0).

1.53 Register RX Frame Counter Port y (0x0320+y*4:0x321+y*4)

Bit	Description	ECAT	PDI	Reset Value
15:0	RX Frame Counter of Port y.	r/ w(clr)	r/-	0

1.54 Register TX Frame Counter Port y (0x0320+y*4:0x321+y*4)

Bit	Description	ECAT	PDI	Reset Value
15:0	TX Frame counter of Port y.	r/ w(clr)	r/-	0

NOTE: Frame Counters 0x0320-0x032F are cleared if one of the RX/TX Frame counters 0x0320-0x032F is written. Write value is ignored (write 0).

1.55 Register Watchdog Divider (0x0400:0x0401)

Bit	Description	ECAT	PDI	Reset Value
15:0	Watchdog divider: Number of 25 MHz tics (minus 2) that represents the basic watchdog increment. (Default value is 100µs = 2498)	r/w	r/-	0x09C2

1.56 Register Watchdog Time PDI (0x0410:0x0411)

Bit	Description	ECAT	PDI	Reset Value
15:0	Watchdog Time PDI: number or basic watchdog increments (Default value with Watchdog divider 100 µs means 100ms Watchdog)	r/w	r/-	0x03E8

1.57 Register Watchdog Time Process Data (0x0420:0x0421)

Bit	Description	ECAT	PDI	Reset Value
15:0	Watchdog Time Process Data: number of basic watchdog increments (Default value with Watchdog divider 100 µs means 100ms Watchdog)	r/w	r/-	0x03E8

There is one Watchdog for all SyncManagers. Watchdog is disabled if Watchdog time is set to 0x0000. Watchdog is restarted with every write access to SyncManagers with Watchdog Trigger Enable Bit set.

1.58 Register Watchdog Status Process Data (0x0440:0x0441)

Bit	Description	ECAT	PDI	Reset Value
0	Watchdog Status of Process Data (triggered by SyncManagers) 0: Watchdog Process Data expired 1: Watchdog Process Data is active or disabled	r/-	r/	0
15:1	Reserved	r/-	r/	0

There is one Watchdog for all SyncManagers. Watchdog is disabled if Watchdog time is set to 0x0000. Watchdog is restarted with every write access to SyncManagers with Watchdog Trigger Enable Bit set.

1.59 Register Watchdog Counter Process Data (0x0442)

Bit	Description	ECAT	PDI	Reset Value
7:0	Watchdog Counter Process Data (counting is stopped when 0xFF is reached). Counts if Process Data Watchdog expires.	r/ w(clr)	r/-	0

NOTE: Watchdog Counters 0x0442-0x0443 are cleared if one of the Watchdog Counters 0x0442-0x0443 is written. Write value is ignored (write 0).

1.60 Register Watchdog Counter PDI (0x0443)

Bit	Description	ECAT	PDI	Reset Value
7:0	Watchdog PDI counter (counting is stopped when 0xFF is reached). Counts if PDI Watchdog expires.	r/ w(clr)	r/-	0

NOTE: Watchdog Counters 0x0442-0x0443 are cleared if one of the Watchdog Counters 0x0442-0x0443 is written. Write value is ignored (write 0).

1.61 Register EEPROM Configuration (0x0500)

Bit	Description	ECAT	PDI	Reset Value
0	EEPROM control is offered to PDI: 0: no 1: yes (PDI has EEPROM control)	r/w	r/-	0
1	Force ECAT access: 0: Do not change Bit 501.0 1: Reset Bit 501.0 to 0	r/w	r/-	0
7:2	Reserved, write 0	r/-	r/-	0

1.62 Register EEPROM PDI Access State (0x0501)

Bit	Description	ECAT	PDI	Reset Value
0	Access to EEPROM: 0: PDI releases EEPROM access 1: PDI takes EEPROM access (PDI has EEPROM control)	r/-	r/(w)	0
7:1	Reserved, write 0	r/-	r/-	0

NOTE: r/(w): write access is only possible if 0x0500.0=1 and 0x0500.1=0.

1.63 Register EEPROM Control/Status (0x0502:0x0503)

Bit	Description	ECAT	PDI	Reset Value
0	ECAT write enable* ¹ : 0: Write requests are disabled 1: Write requests are enabled This bit is always 1 if PDI has EEPROM control.	r/(w)	r/-	0
4:1	Reserved, write 0	r/-	r/-	0
5	EEPROM emulation: 0: Normal operation (I ² C interface used) 1: PDI emulates EEPROM (I ² C not used)	r/-	r/-	0
6	Supported number of EEPROM read bytes: 0: 4 Bytes 1: 8 Bytes	r/-	r/-	1
7	Selected EEPROM Algorithm: 0: 1 address byte (1KBit – 16KBit EEPROMs) 1: 2 address bytes (32KBit – 4 MBit EEPROMs)	r/-	r/-	PIN EEPROM size
10:8	Command register* ¹ : Write: Initiate command. Read: Currently executed command Commands: 000: No command/EEPROM idle (clear error bits) 001: Read 010: Write 100: Reload Others: Reserved/invalid commands (do not issue) EEPROM emulation only: after execution, PDI writes command value to indicate operation is ready.	r/(w)	r/(w)	0
11	Checksum Error at in ESC Configuration Area: 0: Checksum ok 1: Checksum error	r/-	r/-	0
12	EEPROM loading status: 0: EEPROM loaded, device information ok 1: EEPROM not loaded, device information not available (EEPROM loading in progress or finished with a failure)	r/-	r/-	0
13	Error Acknowledge/Command* ² : 0: No error 1: Missing EEPROM acknowledge or invalid command EEPROM emulation only: PDI writes 1 if a temporary failure has occurred.	r/-	r/-	0
14	Error Write Enable* ² : 0: No error 1: Write Command without Write enable	r/-	r/-	0
15	Busy: 0: EEPROM Interface is idle 1: EEPROM Interface is busy	r/-	r/-	0

NOTE: r/(w): write access depends upon the assignment of the EEPROM interface (ECAT/PDI). Write access is generally blocked if EEPROM interface is busy (0x0502.15=1).

*1 Write Enable bit 0 is self-clearing at the SOF of the next frame, Command bits [10:8] are self-clearing after the command is executed (EEPROM Busy ends). Writing "000" to the command register will also clear the error bits [14:13]. Command bits [10:8] are ignored if Error Acknowledge/Command is pending (bit 13).

*2 Error bits are cleared by writing "000" (or any valid command) to Command Register Bits [10:8].

1.64 Register EEPROM Address (0x0504:0x0507)

Bit	Description	ECAT	PDI	Reset Value
31:0	EEPROM Address 0: First word (= 16 bit) 1: Second word ... Actually used EEPROM Address bits: [9:0]: EEPROM size up to 16 kBit [17:0]: EEPROM size 32 kBit – 4 Mbit	r/(w)	r/(w)	0

NOTE: r/(w): write access depends upon the assignment of the EEPROM interface (ECAT/PDI). Write access is generally blocked if EEPROM interface is busy (0x0502.15=1).

1.65 Register EEPROM Data (0x0508:0x050F [0x0508:0x050B])

Bit	Description	ECAT	PDI	Reset Value
15:0	EEPROM Write data (data to be written to EEPROM) or EEPROM Read data (data read from EEPROM, lower bytes)	r/(w)	r/(w)	0
63:16	EEPROM Read data (data read from EEPROM, higher bytes)	r/-	r/-	0

NOTE: r/(w): write access depends upon the assignment of the EEPROM interface (ECAT/PDI). Write access is generally blocked if EEPROM interface is busy (0x0502.15=1).

1.66 Register MII Management Control/Status (0x0510:0x0511)

Bit	Description	ECAT	PDI	Reset Value
-----	-------------	------	-----	-------------

0	Write enable*: 0: Write disabled 1: Write enabled This bit is always 1 if PDI has MI control. ET1100-0000/-0001 exception: Bit is not always 1 if PDI has MI control, and bit is writable by PDI.	r/(w)	r/-	0
1	Management Interface can be controlled by PDI (registers 0x0516-0x0517): 0: Only ECAT control 1: PDI control possible	r/-	r/-	1
2	MI link detection (link configuration, link detection, registers 0x0518-0x051B): 0: Not available 1: MI link detection active	r/-	r/-	0
7:3	PHY address of port 0 PHY address of port 0-3 = PHY_ADOFF+PHY Address.	r/-	r/-	Depends on configuration
9:8	Command register*: Write: Initiate command. Read: Currently executed command Commands: 00: No command/MI idle (clear error bits) 01: Read 10: Write Others: Reserved/invalid commands (do not issue)	r/(w)	r/(w)	0
12:10	Reserved, write 0	r/-	r/-	0
13	Read error: 0: No read error 1: Read error occurred (PHY or register not available) Cleared by writing to this register.	r/(w)	r/(w)	0
14	Command error: 0: Last Command was successful 1: Invalid command or write command without Write Enable Cleared with a valid command or by writing "00" to Command register bits [9:8].	r/-	r/-	0
15	Busy: 0: MI control state machine is idle 1: MI control state machine is active	r/-	r/-	0

NOTE: r/ (w): write access depends on assignment of MI (ECAT/PDI). Write access is generally blocked if Management interface is busy (0x0510.15=1).

* Write enable bit 0 is self-clearing at the SOF of the next frame (or at the end of the PDI access), Command bits [9:8] are self-clearing after the command is executed (Busy ends). Writing "00" to the command register will also clear the error bits [14:13]. The Command bits are cleared after the command is executed.

1.67 Register MII Management Control/Status (0x0510:0x0511)

Bit	Description	ECAT	PDI	Reset Value
0	Write Enable 0: Write Disabled 1: Write Enabled Note: This bit is always 1 if PDI has MI control. (See Note 32)	r/(w)	r/-	0b
1	Management Interface Control 0: ECAT control only 1: MPDI control possible (MII Management ECAT Access State Register and MII Management PDI Access State Register)	r/-	r/-	1b
2	MI Link Detection (Link configuration, link detection, registers PHY Port Status Registers) 0: Not available 1: MI Link Detection Active	r/-	r/-	0b
7:3	PHY Address Offset	r/-	r/-	00000b
9:8	Command Register Write: Initiate command. Read: Currently executed command See Note 30. Commands: 00: No command / MI Idle (clear error bits) 01: Read 10: Write 11: RESERVED (do not issue)	r/(w)	r/(w)	00b
12:10	RESERVED	r/-	r/-	000b
13	Read Error 0: No read error 1: Read error occurred (PHY or register bi available) Note: Cleared by writing this register.	r/(w)	r/(w)	0b
14	Command Error 0: Last command was successful	r/-	r/-	0b

	1: Invalid command or write command without write enable Note: Cleared with a valid command or by writing "00" to Command Register.			
15	Busy 0: MII control state machine is idle 1: MII control state machine is active	r/-	r/-	0b

Note : Write access depends upon the assignment of the MII interface (ECAT/PDI). Write access is generally blocked if the MII interface is busy (Busy bit of MII Management Control/Status Register = 1)

1.68 Register PHY Address (0x0512)

Bit	Description	ECAT	PDI	Reset Value
4:0	PHY Address	r/(w)	r/(w)	0
7:5	Reserved, write 0	r/-	r/-	0

NOTE: r/ (w): write access depends on assignment of MII (ECAT/PDI). Write access is generally blocked if Management interface is busy (0x0510.15=1).

1.69 Register PHY Register Address (0x0513)

Bit	Description	ECAT	PDI	Reset Value
4:0	Address of PHY Register that shall be read/written	r/(w)	r/(w)	0
7:5	Reserved, write 0	r/-	r/-	0

NOTE: r/ (w): write access depends on assignment of MII (ECAT/PDI). Write access is generally blocked if Management interface is busy (0x0510.15=1).

1.70 Register PHY Data (0x0514:0x0515)

Bit	Description	ECAT	PDI	Reset Value
15:0	PHY Read/Write Data	r/(w)	r/(w)	0

NOTE: r/ (w): write access depends on assignment of MII (ECAT/PDI). Access is generally blocked if Management interface is busy (0x0510.15=1).

1.71 Register MII Management ECAT Access State (0x0516)

Bit	Description	ECAT	PDI	Reset Value
0	Access to MII management: 0: ECAT enables PDI takeover of MII management control 1: ECAT claims exclusive access to MII management	r/(w)	r/-	0
7:1	Reserved, write 0	r/-	r/-	0

NOTE: r/ (w): write access is only possible if 0x0517.0=0.

1.72 Register MII Management PDI Access State (0x0517)

Bit	Description	ECAT	PDI	Reset Value
0	Access to MII management: 0: ECAT has access to MII management 1: PDI has access to MII management	r/-	r/(w)	0
1	Force PDI Access State: 0: Do not change Bit 517.0 1: Reset Bit 517.0 to 0	r/w	r/-	0
7:2	Reserved, write 0	r/-	r/-	0

NOTE: r/ (w): assigning access to PDI (bit 0 = 1) is only possible if 0x0516.0=0 and 0x0517.1=0, and if the SII EEPROM is loaded (0x0110[0]=1).

1.73 Register Logical Start address FMMU y (0x06y0:0x06y3)

Bit	Description	ECAT	PDI	Reset Value
31:0	Logical start address within the EtherCAT Address Space.	r/w	r/-	0

1.74 Register Length FMMU y (0x06y4:0x06y5)

Bit	Description	ECAT	PDI	Reset Value
15:0	Offset from the first logical FMMU Byte to the last FMMU Byte + 1 (e.g., if two bytes are used then this parameter shall contain 2)	r/w	r/-	0

1.75 Register Start bit FMMU y in logical address space (0x06y6)

Bit	Description	ECAT	PDI	Reset Value
2:0	Logical starting bit that shall be mapped (bits are counted from least significant bit (=0) to most significant bit(=7)	r/w	r/-	0
7:3	Reserved, write 0	r/-	r/-	0

1.76 Register Stop bit FMMU y in logical address space (0x06y7)

Bit	Description	ECAT	PDI	Reset Value
2:0	Last logical bit that shall be mapped (bits are counted from least significant bit (=0) to most significant bit(=7)	r/w	r/-	0
7:3	Reserved, write 0	r/-	r/-	0

1.77 Register Physical Start address FMMU y (0x06y8-0x06y9)

Bit	Description	ECAT	PDI	Reset Value
-----	-------------	------	-----	-------------

15:0	Physical Start Address (mapped to logical Start address)	r/w	r/-	0
------	--	-----	-----	---

1.78 Register Physical Start bit FMMU y (0x06yA)

Bit	Description	ECAT	PDI	Reset Value
2:0	Physical starting bit as target of logical start bit mapping (bits are counted from least significant bit (=0) to most significant bit(=7)	r/w	r/-	0
7:3	Reserved, write 0	r/-	r/-	0

1.79 Register Type FMMU y (0x06yB)

Bit	Description	ECAT	PDI	Reset Value
0	0: Ignore mapping for read accesses 1: Use mapping for read accesses	r/w	r/-	0
1	0: Ignore mapping for write accesses 1: Use mapping for write accesses	r/w	r/-	0
7:2	Reserved, write 0	r/-	r/-	0

1.80 Register Activate FMMU y (0x06yC)

Bit	Description	ECAT	PDI	Reset Value
0	0: FMMU deactivated 1: FMMU activated. FMMU checks logical addressed blocks to be mapped according to mapping configured	r/w	r/-	0
7:1	Reserved, write 0	r/-	r/-	0

1.81 Register Reserved FMMU y (0x06yD:0x06yF)

Bit	Description	ECAT	PDI	Reset Value
23:0	Reserved, write 0	r/-	r/-	0

1.82 Register physical Start Address SyncManager y (0x0800+y*8:0x0801+y*8)

Bit	Description	ECAT	PDI	Reset Value
15:0	Specifies first byte that will be handled by SyncManager	r/(w)	r/-	0

NOTE r/(w): Register can only be written if SyncManager is disabled (+0x6.0 = 0).

1.83 Register Length SyncManager y (0x0802+y*8:0x0803+y*8)

Bit	Description	ECAT	PDI	Reset Value
15:0	Number of bytes assigned to SyncManager (shall be greater 1, otherwise SyncManager is not activated. If set to 1, only Watchdog Trigger is generated if configured)	r/(w)	r/-	0

NOTE r/(w): Register can only be written if SyncManager is disabled (+0x6.0 = 0).

1.84 Register Control Register SyncManager y (0x0804+y*8)

Bit	Description	ECAT	PDI	Reset Value
1:0	Operation Mode: 00: Buffered (3 buffer mode) 01: Reserved 10: Mailbox (Single buffer mode) 11: Reserved	r/(w)	r/-	0
3:2	Direction: 00: Read: ECAT read access, PDI write access. 01: Write: ECAT write access, PDI read access. 10: Reserved 11: Reserved	r/(w)	r/-	0
4	Interrupt in ECAT Event Request Register: 0: Disabled 1: Enabled	r/(w)	r/-	0
5	Interrupt in PDI Event Request Register: 0: Disabled 1: Enabled	r/(w)	r/-	0
6	Watchdog Trigger Enable: 0: Disabled 1: Enabled	r/(w)	r/-	0
7	Reserved, write 0	r/-	r/-	0

NOTE r/(w): Register can only be written if SyncManager is disabled (+0x6.0 = 0).

1.85 Register Status Register SyncManager y (0x0805+y*8)

Bit	Description	ECAT	PDI	Reset Value
0	Interrupt Write: 1: Interrupt after buffer was completely and successfully written 0: Interrupt cleared after first byte of buffer was read NOTE: This interrupt is signaled to the reading side if enabled in the SM Control	r/-	r/-	0

1	Interrupt Read: 1: Interrupt after buffer was completely and successful read 0: Interrupt cleared after first byte of buffer was written NOTE: This interrupt is signaled to the writing side if enabled in the SM Control register.	r/-	r/-	0
2	Reserved	r/-	r/-	0
3	Mailbox mode: mailbox status: 0: Mailbox empty 1: Mailbox full Buffered mode: reserved	r/-	r/-	0
5:4	Buffered mode: buffer status (last written buffer): 00: 1. buffer 01: 2. buffer 10: 3. buffer 11: (no buffer written) Mailbox mode: reserved	r/-	r/-	11
6	Read buffer in use (opened)	r/-	r/-	0
7	Write buffer in use (opened)	r/-	r/-	0

1.86 Register Activate SyncManager y (0x0806+y*8)

Bit	Description	ECAT	PDI	Reset Value
0	SyncManager Enable/Disable: 0: Disable: Access to Memory without SyncManager control 1: Enable: SyncManager is active and controls Memory area set in configuration	r/w	r/	0
1	Repeat Request: A toggle of Repeat Request means that a mailbox retry is needed (primarily used in conjunction with ECAT Read Mailbox)	r/w	r/	0
5:2	Reserved, write 0	r/-	r/	0
6	Latch Event ECAT: 0: No 1: Generate Latch event if EtherCAT master issues a buffer exchange	r/w	r/	0
7	Latch Event PDI: 0: No 1: Generate Latch events if PDI issues a buffer exchange or if PDI accesses buffer start address	r/w	r/	0

PDI register function acknowledge by Write command is enabled: Writing this register from PDI in all SyncManagers which have changed activation clears AL Event Request 0x0220[4]. Writing to this register from PDI is possible; write value is ignored (write 0).

1.87 Register PDI Control SyncManager y (0x0807+y*8)

Bit	Description	ECAT	PDI	Reset Value
0	Read: 0: Normal operation, SyncManager activated 1: SyncManager deactivated and reset SyncManager locks access to Memory area. Write: 0: Activate SyncManager 1: Request SyncManager deactivation NOTE: Writing 1 is delayed until the end of a frame which is currently processed.	r/-	r/w	0
1	Repeat Ack: If this is set to the same value as set by Repeat Request, the PDI acknowledges the execution of a previous set Repeat request.	r/-	r/w	0
7:2	Reserved, write 0	r/-	r/-	0

1.88 Register Receive Time Port 0 (0x0900:0x0903)

Bit	Description	ECAT	PDI	Reset Value
31:0	Write: A write access to register 0x0900 with BWR or FPWR latches the local time of the beginning of the receive frame (start first bit of preamble) at each port. Write: A write access latches the local time of the beginning of the receive frame at port 0. It enables the time stamping at the other ports. Read: Local time of the beginning of the last receive frame containing a write access to this register.	r/w (special function)	r/-	Undefined

NOTE: The time stamps cannot be read in the same frame in which this register was written.

1.89 Register Receive Time Port 1 (0x0904:0x0907)

Bit	Description	ECAT	PDI	Reset Value
31:0	Local time of the beginning of a frame (start first bit of preamble) received at port 1 containing a BWR or FPWR to Register 0x0900. Local time of the beginning of the first frame received at port 1 after time stamping was enabled. Time stamping is disabled for this port afterwards.	r/-	r/-	Undefined

1.90 Register Receive Time Port 2 (0x0908:0x090B)

Bit	Description	ECAT	PDI	Reset Value
31:0	Local time of the beginning of a frame (start first bit of preamble) received at port 2 containing a BWR or FPWR to Register 0x0900.	r/-	r/-	Undefined

1.91 Register Receive Time Port 3 (0x090C:0x090F)

Bit	Description	ECAT	PDI	Reset Value
31:0	Local time of the beginning of a frame (start first bit of preamble) received at port 3 containing a BWR or FPWR to Register 0x0900. Local time of the beginning of the first frame received at port 3 after time stamping was enabled. Time stamping is disabled for this port afterwards.	r/-	r/-	Undefined

1.92 Register Receive Time ECAT Processing Unit (0x0918:0x091F)

Bit	Description	ECAT	PDI	Reset Value
63:0	Local time of the beginning of a frame (start first bit of preamble) received at the ECAT Processing Unit containing a write access to Register 0x0900 Local time of the beginning of the frame received at the ECAT Processing Unit containing a write access to register 0x0900:0x0903. NOTE: E.g., if port 0 is open, this register reflects the Receive Time Port 0 as a 64 Bit value.	r/-	r/-	Undefined

1.93 Register System Time (0x0910:0x0913 [0x0910:0x0917])

Bit	Description	ECAT	PDI	Reset Value
63:0	ECAT read access: Local copy of the System Time when the frame passed the reference clock (i.e., including System Time Delay). Time latched at beginning of the frame (Ethernet SOF delimiter)	r	-	0
63:0	PDI read access: Local copy of the System Time. Time latched when reading first byte (0x0910)	-	r	
31:0	Write access: Written value will be compared with the local copy of the System time. The result is an input to the time control loop. NOTE: written value will be compared at the end of the frame with the latched (SOF) local copy of the System time if at least the first byte (0x0910) was written.	(w) (special function)	-	
31:0	Write access: Written value will be compared with Latch0 Time Positive Edge time. The result is an input to the time control loop. NOTE: written value will be compared at the end of the access with Latch0 Time Positive Edge (0x09B0:0x09B3) if at least the last byte (0x0913) was written.	-	(w) (special function)	

NOTE: Write access to this register depends upon ESC configuration (typically ECAT, PDI only with explicit ESC configuration: System Time PDI controlled).

1.94 Register System Time Offset (0x0920:0x0927)

Bit	Description	ECAT	PDI	Reset Value
63:0	Difference between local time and System Time. Offset is added to the local time.	r/(w)	r/(w)	0

NOTE: Write access to this register depends upon ESC configuration (typically ECAT, PDI only with explicit ESC configuration: System Time PDI controlled). Reset internal system time difference filter and speed counter filter by writing Speed Counter Start (0x0930:0x0931) after changing this value.

1.95 Register System Time Delay (0x0928:0x092B)

Bit	Description	ECAT	PDI	Reset Value
31:0	Delay between Reference Clock and the ESC	r/(w)	r/(w)	0

NOTE: Write access to this register depends upon ESC configuration (typically ECAT, PDI only with explicit ESC configuration: System Time PDI controlled). Reset internal system time difference filter and speed counter filter by writing Speed Counter Start (0x0930:0x0931) after changing this value.

1.96 Register System Time Difference (0x092C:0x092F)

Bit	Description	ECAT	PDI	Reset Value
30:0	Mean difference between local copy of System Time and received System Time values	r/-	r/-	0
31	0: Local copy of System Time greater than or equal received System Time values 1: Local copy of System Time smaller than received System Time	r/-	r/-	0

1.97 Register Speed Counter Start (0x0930:0x0931)

Bit	Description	ECAT	PDI	Reset Value
14:0	Bandwidth for adjustment of local copy of System Time (larger values → smaller bandwidth and smoother adjustment) A write access resets System Time Difference (0x092C:0x092F) and Speed Counter Diff (0x0932:0x0933). Counter Diff (0x0932:0x0933). Minimum value: 0x0080 to 0x3FFF	r/(w)	r/(w)	0x1000
15	Reserved, write 0	r/-	r/-	0

NOTE: Write access to this register depends upon ESC configuration (typically ECAT, PDI only with explicit ESC configuration: System Time PDI controlled).

1.98 Register Speed Counter Diff (0x0932:0x0933)

Bit	Description	ECAT	PDI	Reset Value
15:0	Representation of the deviation between local clock period and Reference Clock's clock period (representation: two's complement) Range: $\pm(\text{Speed Counter Start} - 0x7F)$	r/-	r/-	0x0000

NOTE: Calculate the clock deviation after System Time Difference has settled at a low value as follows:
Deviation: $\text{Speed Counter Diff} / (5(\text{Speed Counter Start} + \text{Speed Counter Diff} + 2)(\text{Speed Counter Start} - \text{Speed Counter Diff} + 2))$

1.99 Register System Time Difference Filter Depth (0x0934)

Bit	Description	ECAT	PDI	Reset Value
3:0	Filter depth for averaging the received System Time deviation. A write access resets System Time Difference (0x092C:0x092F)	r/(w)	r/(w)	4
7:4	Reserved, write 0	r/-	r/-	0

NOTE: Write access to this register depends upon ESC configuration (typically ECAT, PDI only with explicit ESC configuration: System Time PDI controlled).

1.100 Register Speed Counter Filter Depth (0x0935)

Bit	Description	ECAT	PDI	Reset Value
3:0	Filter depth for averaging the clock period deviation A write access resets the internal speed counter filter.	r/(w)	r/(w)	12
7:4	Reserved, write 0	r/-	r/-	0

NOTE: Write access to this register depends upon ESC configuration (typically ECAT, PDI only with explicit ESC configuration: System Time PDI controlled).

1.101 Register Receive Time Latch Mode (0x0936)

Bit	Description	ECAT	PDI	Reset Value
0	Receive Time Latch Mode: 0: Forwarding mode (used if frames are entering the ESC at port 0 first): Receive time stamps of ports 1-3 are enabled after the write access to 0x0900, so the following frame at ports 1-3 will be time stamped (this is typically the write frame to 0x0900 coming back from the network behind the ESC). 1: Reverse mode (used if frames are entering ESC at port 1-3 first): Receive time stamps of ports 1-3 are immediately taken over from the internal hidden time stamp registers, so the previous frame entering the ESC at ports 1-3 will be time stamped when the write frame to 0x0900 enters port 0 (the previous frame at ports 1-3 is typically the write frame to 0x0900 coming from the master, which will enable time stamping at the ESC once it enters port 0).	r/w	r/-	0
7:1	Reserved	r/-	r/-	0

NOTE: There should not be frames traveling around the network before and after the time stamps are taken, otherwise these frames might get time stamped, and not the write frame to 0x0900.

1.102 Register Cyclic Unit Control (0x0980)

Bit	Description	ECAT	PDI	Reset Value
0	SYNC out unit control: 0: ECAT controlled 1: PDI controlled	r/w	r/-	0
3:1	Reserved, write 0	r/-	r/-	0
4	Latch In unit 0: 0: ECAT controlled 1: PDI controlled NOTE: Always 1 (PDI controlled) if System Time is PDI controlled. Latch interrupt is routed to ECAT/PDI depending on this setting	r/w	r/-	0
5	Latch In unit 1: 0: ECAT controlled 1: PDI controlled NOTE: Latch interrupt is routed to ECAT/PDI depending on this setting	r/w	r/-	0
7:6	Reserved, write 0	r/-	r/-	0

1.103 Register Activation register (0x0981)

Bit	Description	ECAT	PDI	Reset Value
0	Sync Out Unit activation: 0: Deactivated 1: Activated	r/(w)	r/(w)	0
1	SYNC0 generation: 0: Deactivated 1: SYNC0 pulse is generated	r/(w)	r/(w)	0
2	SYNC1 generation: 0: Deactivated 1: SYNC1 pulse is generated	r/(w)	r/(w)	0
3	Auto-activation by writing Start Time Cyclic Operation (0x0990:0x0997): 0: Disabled 1: Auto-activation enabled. 0x0981.0 is set automatically after Start Time is written.	r/(w)	r/(w)	0
4	Extension of Start Time Cyclic Operation (0x0990:0x0993): 0: No extension 1: Extend 32 bit written Start Time to 64 bit	r/(w)	r/(w)	0
5	Start Time plausibility check: 0: Disabled. SyncSignal generation if Start Time is reached. 1: Immediate SyncSignal generation if Start Time is outside near future (see 0x0981.6)	r/(w)	r/(w)	0

6	Near future configuration (approx.): 0: ½ DC width future (231 ns or 263 ns) 1: ~2.1 sec. future (231 ns)	r/(w)	r/(w)	0
7	SyncSignal debug pulse (Vasily bit): 0: Deactivated 1: Immediately generate one ping only on SYNC0-1 according to 0x0981[2:1] for debugging This bit is self-clearing, always read 0.	r/(w)	r/(w)	0

NOTE: Write to this register depends upon setting of 0x0980.0.

1.104 Register Pulse Length of SyncSignals (0x0982:0x983)

Bit	Description	ECAT	PDI	Reset Value
15:0	Pulse length of SyncSignals (in Units of 10ns) 0: Acknowledge mode: SyncSignal will be cleared by reading SYNC[1:0] Status register	r/-	r/-	0, later EEPROM ADR 0x0002

1.105 Register Pulse Length of SyncSignals (0x0982:0x983)

Bit	Description	ECAT	PDI	Reset Value
15:0	Pulse length of SyncSignals (in units of 10ns) A value of 0 is used for Acknowledge Mode: SyncSignal will Be cleared by reading the SYNC0 Status Register/SYNC1 Status Register.	r/-	r/-	0000h

Note: The default value of this field can be configured via EEPROM.

Note: For EtherCAT Core CSR registers longer than one byte, the LSB has the lowest address and the MSB the highest address.

1.106 Register Activation Status (0x0984)

Bit	Description	ECAT	PDI	Reset Value
0	SYNC0 activation state: 0: First SYNC0 pulse is not pending 1: First SYNC0 pulse is pending	r/-	r/-	0
1	SYNC1 activation state: 0: First SYNC1 pulse is not pending 1: First SYNC1 pulse is pending	r/-	r/-	0
2	Start Time Cyclic Operation (0x0990:0x0997) plausibility check result when Sync Out Unit was activated: 0: Start Time was within near future 1: Start Time was out of near future (0x0981.6)	r/-	r/-	0
7:3	Reserved	r/-	r/-	0

1.107 Register SYNC0 Status (0x098E)

Bit	Description	ECAT	PDI	Reset Value
0	SYNC0 state for Acknowledge mode. SYNC0 in Acknowledge mode is cleared by reading this register from PDI, use only in Acknowledge mode	r/-	r/	0
7:1	Reserved	r/-	r/	0

* Reading this register from PDI clears AL Event Request 0x0220[2].

1.108 Register SYNC1 Status (0x098F)

Bit	Description	ECAT	PDI	Reset Value
0	SYNC1 state for Acknowledge mode. SYNC1 in Acknowledge mode is cleared by reading this register from PDI, use only in Acknowledge mode	r/-	r/	0
7:1	Reserved	r/-	r/	0

Reading this register from PDI clears AL Event Request 0x0220[3].

1.109 Register Start Time Cyclic Operation (0x0990:0x0993 [0x0990:0x0997])

Bit	Description	ECAT	PDI	Reset Value
63:0	Write: Start time (System time) of cyclic operation in ns Read: System time of next SYNC0 pulse in ns	r/(w)	r/(w)	0

NOTE: Write to this register depends upon setting of 0x0980.0. Only writable if 0x0981.0=0. Auto-activation (0x0981.3=1): upper 32 bits are automatically extended if only lower 32 bits are written within one frame.

1.110 Register Next SYNC1 Pulse (0x0998:0x099B [0x0998:0x099F])

Bit	Description	ECAT	PDI	Reset Value
63:0	System time of next SYNC1 pulse in ns	r/-	r/-	0

1.111 Register SYNC0 Cycle Time (0x09A0:0x09A3)

Bit	Description	ECAT	PDI	Reset Value
31:0	Time between two consecutive SYNC0 pulses in ns. 0: Single shot mode, generate only one SYNC0 pulse.	r/(w)	r/(w)	0

NOTE: Write to this register depends upon setting of 0x0980.0.

1.112 Register SYNC1 Cycle Time (0x09A4:0x09A7)

Bit	Description	ECAT	PDI	Reset Value
31:0	Time between SYNC1 pulses and SYNC0 pulse in ns	r/(w)	r/(w)	0

NOTE: Write to this register depends upon setting of 0x0980.0.

1.113 Register Latch0 Control (0x09A8)

Bit	Description	ECAT	PDI	Reset Value
0	Latch0 positive edge: 0: Continuous Latch active 1: Single event (only first event active)	r/(w)	r/(w)	0
1	Latch0 negative edge: 0: Continuous Latch active 1: Single event (only first event active)	r/(w)	r/(w)	0
7:2	Reserved, write 0	r/-	r/-	0

NOTE: Write access depends upon setting of 0x0980.4.

1.114 Register Latch1 Control (0x09A9)

Bit	Description	ECAT	PDI	Reset Value
0	Latch1 positive edge: 0: Continuous Latch active 1: Single event (only first event active)	r/(w)	r/(w)	0
1	Latch1 negative edge: 0: Continuous Latch active 1: Single event (only first event active)	r/(w)	r/(w)	0
7:2	Reserved, write 0	r/-	r/-	0

NOTE: Write access depends upon setting of 0x0980.5.

1.115 Register Latch0 Status (0x09AE)

Bit	Description	ECAT	PDI	Reset Value
0	Event Latch0 positive edge. 0: Positive edge not detected or continuous mode 1: Positive edge detected in single event mode only. Flag cleared by reading out Latch0 Time Positive Edge.	r/-	r/-	0
1	Event Latch0 negative edge. 0: Negative edge not detected or continuous mode 1: Negative edge detected in single event mode only. Flag cleared by reading out Latch0 Time Negative Edge.	r/-	r/-	0
2	Latch0 pin state	r/-	r/-	0
7:3	Reserved	r/-	r/-	0

1.116 Register Latch1 Status (0x09AF)

Bit	Description	ECAT	PDI	Reset Value
0	Event Latch1 positive edge. 0: Positive edge not detected or continuous mode 1: Positive edge detected in single event mode only. Flag cleared by reading out Latch1 Time Positive Edge.	r/-	r/-	0
1	Event Latch1 negative edge. 0: Negative edge not detected or continuous mode 1: Negative edge detected in single event mode only. Flag cleared by reading out Latch1 Time Negative Edge.	r/-	r/-	0
2	Latch1 pin state	r/-	r/-	0
7:3	Reserved	r/-	r/-	0

1.117 Register Latch0 Time Positive Edge (0x09B0:0x09B7)

Bit	Description	ECAT	PDI	Reset Value
63:0	Register captures System time at the positive edge of the Latch0 signal.	r(ack)/-	r/	0

NOTE: Register bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value. Reading this register from ECAT clears Latch0 Status 0x09AE[0] if 0x0980.4=0. Writing to this register from ECAT is not possible.

* Reading this register from PDI if 0x0980.4=1 clears Latch0 Status 0x09AE[0].

1.118 Register Latch0 Time Negative Edge (0x09B8:0x09BF)

Bit	Description	ECAT	PDI	Reset Value
63:0	Register captures System time at the negative edge of the Latch0 signal.	r/-	r/	0

NOTE: Register bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value. Reading this register from ECAT clears Latch0 Status 0x09AE[1] if 0x0980.4=0. Writing to this register from ECAT is not possible.

* Reading this register from PDI if 0x0980.4=1 clears Latch0 Status 0x09AE[1].

1.119 Register Latch1 Time Positive Edge (0x09C0:0x09C7)

Bit	Description	ECAT	PDI	Reset Value
63:0	Register captures System time at the positive edge of the Latch1 signal.	r/-	r/	0

NOTE: Register bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value. Reading this register from ECAT clears Latch0 Status 0x09AF[0] if 0x0980.5=0. Writing to this register from ECAT is not possible.

1.120 Register Latch1 Time Negative Edge (0x09C8:0x09CF)

Bit	Description	ECAT	PDI	Reset Value
63:0	Register captures System time at the negative edge of the Latch1 signal.	r/-	r/	0

NOTE: Register bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value. Reading this register from ECAT clears Latch0 Status 0x09AF[1] if 0x0980.5=0. Writing to this register from ECAT is not possible.

* Reading this register from PDI if 0x0980.5=1 clears Latch0 Status 0x09AF[1]. Writing to this register from PDI is not possible.

1.121 Register EtherCAT Buffer Change Event Time (0x09F0:0x09F3)

Bit	Description	ECAT	PDI	Reset Value
31:0	Register captures local time of the beginning of the frame which causes at least one SyncManager to assert an ECAT event	r/-	r/-	0

NOTE: Register bits [31:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value.

1.122 Register PDI Buffer Start Event Time (0x09F8:0x09FB)

Bit	Description	ECAT	PDI	Reset Value
31:0	Register captures local time when at least one SyncManager asserts an PDI buffer start event	r/-	r/-	0

NOTE: Register bits [31:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value.

1.123 Register PDI Buffer Change Event Time (0x09FC:0x09FF)

Bit	Description	ECAT	PDI	Reset Value
31:0	Register captures local time when at least one SyncManager asserts an PDI buffer change event	r/-	r/-	0

NOTE: Register bits [31:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value.

1.124 Register Power-On Values TR8211K (0x0E00:0x0E01)

Bit	Description	ECAT	PDI	Reset Value
1:0	Port mode (P_MODE): 00: Logical ports 0 and 1 available 01: Logical ports 0, 1 and 2 available 10: Logical ports 0, 1 and 3 available 11: Logical ports 0, 1, 2 and 3 available	r/-	r/-	Depends on Hardware configuration
5:2	Physical layer of available ports (P_CONF). Bit 2 → logical port 0, Bit 3 → logical port 1, Bit 4 → third logical port(2/3), Bit 5 → logical port 3. 0: EBUS 1: MII	r/-	r/-	
7:6	CPU clock output (CLK_MODE): 00: Off – PDI[7] available as PDI port 01: PDI[7] = 25MHz 10: PDI[7] = 20MHz 11: PDI[7] = 10MHz	r/-	r/-	
9:8	TX signal shift (C25_SHI): 00: MII TX signals shifted by 0°	r/-	r/-	

	01: MII TX signals shifted by 90° 10: MII TX signals shifted by 180° 11: MII TX signals shifted by 270°			
10	CLK25 Output Enable (C25_ENA): 0: Disabled – PDI[31] available as PDI port 1: Enabled – PDI[31] = 25MHz (OSC)	r/-	r/-	
11	Reserved configuration bit	r/-	r/-	
12	Digital Control/State Move (Ctrl_Status_Move): 0: Control/Status signals are mapped to PDI[39:32] if available 1: Control/Status signals are remapped to the highest available PDI Byte.	r/-	r/-	
13	PHY Address Offset (PHYAD_OFF): 0: No PHY address offset 1: PHY address offset is 16	r/-	r/-	
14	PHY Link Polarity (LINKPOL): 0: LINK_MII is active low 1: LINK_MII is active high	r/-	r/-	
15	Reserved configuration bit	r/-	r/-	

1.125 Register Power-On Values TR8253/TR8253L (0x0E00:0x0E01)

Bit	Description	ECAT	PDI	Reset Value
1:0	Chip mode: 0x: 2 port mode. Port 0 = PHY A, Port 1 = PHY B 10: 3 port downstream mode. Port 0 = PHY A, Port 1 = PHY B, Port 2 = MII 11: 3 port upstream mode. Port 0 = MII, Port 1 = PHY B, Port 2 = PHY A	r/-	r/-	Depends on Hardware configuration
2	NAND of EEPROM_SCL and EEPROM_SDA	r/-	r/-	
3	Selected EEPROM Algorithm: 0: 1 address byte (1KBit – 16KBit EEPROMs) 1: 2 address bytes (32KBit – 4 MBit EEPROMs)	r/-	r/-	
4	Reserved configuration bit	r/-	r/-	
5	Selected EEPROM Algorithm: 0: 1 address byte (1KBit – 16KBit EEPROMs) 1: 2 address bytes (32KBit – 4 MBit EEPROMs)	r/-	r/-	
6	EEPROM_SCL	r/-	r/-	
7	EEPROM_SDA	r/-	r/-	
8	Reserved configuration bit	r/-	r/-	
10:9	TX signal shift (C25_SHI): 00: MII TX signals shifted by 270° 01: MII TX signals shifted by 0° 10: MII TX signals shifted by 90° 11: MII TX signals shifted by 180°	r/-	r/-	
11	PHY Link Polarity (LINKPOL): 0: LINK_MII is active low 1: LINK_MII is active high	r/-	r/-	
12	Reserved configuration bit	r/-	r/-	
13	CLK25 Output Enable (C25_ENA): 0: Disabled 1: Enabled	r/-	r/-	
14	Reserved configuration bit	r/-	r/-	
15	Reserved configuration bit	r/-	r/-	

1.126 Register Vendor ID (0x0E08:0x0E0F)

Bit	Description	ECAT	PDI	Reset Value
31:0	Vendor ID:	r/-	r/-	6500
63:32	Reserved	r/-	r/-	

1.127 Register Digital I/O Output Data (0x0F00:0x0F03)

Bit	Description	ECAT	PDI	Reset Value
31:0	Output Data	r/w	r/-	0

NOTE: Register size depends on PDI setting and/or device configuration. This register is bit-writable (using Logical addressing).

1.128 Register General Purpose Outputs (0x0F10:0x0F17)

Bit	Description	ECAT	PDI	Reset Value
15:0	General Purpose Output Data	r/w	r/w	0

NOTE: Register size depends on PDI setting and/or device configuration.

1.129 Register General Purpose Inputs (0x0F18:0x0F1F)

Bit	Description	ECAT	PDI	Reset Value
15:0	General Purpose Input Data	r/-	r/-	0

NOTE: Register size depends on PDI setting and/or device configuration.

1.130 User RAM (0x0F80:0x0FFF)

Bit	Description	ECAT	PDI	Reset Value
	Application specific information	r/w	r/w	Random/undefined

1.131 Digital I/O Input Data (0x1000:0x1003)

Bit	Description	ECAT	PDI	Reset Value
31:0	Input Data	(r/w)	(r/w)	Random/undefined

NOTE (r/w): Process Data RAM is only accessible if EEPROM was correctly loaded (register 0x0110.0 = 1).

NOTE: Input Data size depends on PDI setting and/or device configuration. Digital I/O Input Data is written into the Process Data RAM at these addresses if a Digital I/O PDI with inputs is configured.

1.132 Process Data RAM (0x1000:0xFFFF)

Bit	Description	ECAT	PDI	Reset Value
	Process Data RAM	(r/w)	(r/w)	Random/undefined

NOTE (r/w): Process Data RAM is only accessible if EEPROM was correctly loaded (register 0x0110.0 = 1).

2.ESC ConfigData

Word Address	Parameter Description	Register Address	Bit	Description
0x0	PDI Control/ESC Configuration: Initialization value for PDI Control register (EEPROM ADR 0x0000[9] is also mapped to register 0x0110[2])	0x0140: 0x0141	[7:0]	Process data interface: 0x00: Interface deactivated (no PDI) 0x04: Digital I/O 0x05: Beckhoff SPI Slave 0x08: 16 Bit asynchronous Microcontroller interface 0x09: 8 Bit asynchronous Microcontroller interface 0x83: Triductor SQI Slave 0x90: HBI Multiplexed 1 Phase 8-bit 0x91: HBI Multiplexed 1 Phase 16-bit 0x92: HBI Multiplexed 2 Phase 8-bit 0x93: HBI Multiplexed 2 Phase 16-bit 0x94: HBI Indexed 8-bit 0x95: HBI Indexed 16-bit 0x96: HBI Demultiplexed 8-bit 0x97: HBI Demultiplexed 16-bit Others: Reserved
			[8]	Device emulation (control of AL status): 0: AL status register has to be set by PDI 1: AL status register will be set to value written to AL control register
			[9]	Enhanced Link detection all ports: 0: disabled (if bits [7:4]=0) 1: enabled at all ports (overrides bits [7:4])
			[10]	Distributed Clocks SYNC Out Unit: 0: disabled (power saving) 1: enabled
			[11]	Distributed Clocks Latch In Unit: 0: disabled (power saving) 1: enabled
			[12]	Enhanced Link port 0: 0: disabled (if bit 9=0) 1: enabled
			[13]	Enhanced Link port 1: 0: disabled (if bit 9=0) 1: enabled
			[14]	Enhanced Link port 2: 0: disabled (if bit 9=0) 1: enabled
			[15]	Enhanced Link port 3: 0: disabled (if bit 9=0) 1: enabled
0x1	PDI Configuration: Initialization value for PDI Configuration register	0x0150 Digital I/O (0x140:0x4)	[0]	OUTVALID polarity: 0: Active high 1: Active low
			[1]	OUTVALID mode: 0: Output event signaling 1: Process Data Watchdog trigger (WD_TRIG) signaling on OUTVALID pin (see SyncManager). Output data is updated if watchdog is triggered. Overrides 0x0150[7:6]
			[2]	Unidirectional/Bidirectional mode: 0: Unidirectional mode: input/output direction of pins configured individually 1: Bidirectional mode: all I/O pins are bidirectional, direction configuration is ignored
			[3]	Watchdog behavior: 0: Outputs are reset immediately after watchdog expires 1: Outputs are reset with next output event that follows watchdog expiration
			[5:4]	Input DATA is sampled at 00: Start of Frame2 01: Rising edge of LATCH_IN 10: DC SYNC0 event2 11: DC SYNC1 event2
			[7:6]	Output DATA is updated at 00: End of Frame 01: Reserved 10: DC SYNC0 event 11: DC SYNC1 event If 0x0150[1]=1, output DATA is updated at Process Data Watchdog trigger event (0x0150[7:6] are ignored)

		0x0150 SPI(0x0140:0x5) SQI(0x0140:0x83)	[1:0]	SPI mode: 00: SPI mode 0 01: SPI mode 1 10: SPI mode 2 11: SPI mode 3 NOTE: SPI mode 3 is recommended for Slave Sample Code NOTE: SPI status flag is not available in SPI modes 0 and 2 with normal data out sample. NOTE: SQI Not Available
			[3:2]	SPI_IRQ/SQI_IRQ output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)
			[4]	Reserved, set EEPROM value 0
			[5]	Data Out sample mode: 0: Normal sample (SPI_DO and SPI_DI are sampled at the same SPI_CLK edge) 1: Late sample (SPI_DO and SPI_DI are sampled at different SPI_CLK edges) NOTE: SQI Not Available
			[7:6]	Reserved, set EEPROM value 0
		0x0150 Oversampling (0x0140:0x6)	0	Oversampling Output Enable Pin Polarity 0: Active Low 1: Active High
			1	Second Channel Enable: 0: Disable 1: Enable
			[3:2]	Oversampling Direction: 01: Input 10: Output Others: Reserved
			[7:4]	Reserved, set EEPROM value 0
		0x0150 HBI (0x140:0x8/0x9)	[1:0]	BUSY output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high) NOTE: Push-Pull: no CS → not BUSY (driven) Open Drain/Source: no CS → BUSY open
			[3:2]	IRQ output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)
			[4]	BHE/Byte Enable polarity: 0: Active low 1: Active high
			[5]	Reserved, set EEPROM value 0
			[6]	Reserved, set EEPROM value 0
			[7]	RD Polarity: 0: Active low 1: Active high
		0x0150 HBI (0x140:0x90~0x97)	[0]	HBI WAIT_ACK Buffer Type 0: Push-Pull 1: Open Drain
			[1]	HBI WAIT_ACK Polarity 0: Active low 1: Active high
			[2]	HBI ALE Polarity Configures the polarity of the HBI interface ALEHI and ALELO signals 0: Active Low Strobe (Address saved on rising edge) 1: Active High Strobe (Address saved on falling edge) NOTE: Only Multiplexed Available
			[3]	HBI Write, Enable Polarity Configures the polarity of the HBI interface write signal. 0: Active Low Write 1: Active High Write Configures the polarity of the HBI interface read/write signal. 0: Active Low Enable 1: Active High Enable
			[4]	HBI Read, Read/Write Polarity Configures the polarity of the HBI interface read signal.

				0: Active Low Read 1: Active High Read Configures the polarity of the HBI interface read/write signal. 0: Read when 1, write when 0 (R/nW) 1: Write when 1, read when 0 (W/nR)
			[5]	HBI Chip Select Polarity Configures the polarity of the HBI interface chip select signal. 0: Active Low 1: Active High
			[6]	HBI Read/Write Mode Configures the HBI interface for separate read and write signals or direction and enable signals. 0: Read and Write 1: Direction and Enable
			[7]	HBI ALE Qualification Configures the HBI interface to qualify the ALEHI and ALELO signals with the CS signal. 0: Addr input is latched with ALEHI and ALELO 1: Addr input is latched with ALEHI and ALELO only when CS is active
		0X0151 Sync/Latch	[8:9]	SYNC0 output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)
			[10]	SYNC0/LATCH0 configuration: 0: LATCH0 Input 1: SYNC0 Output
			[11]	SYNC0 mapped to AL Event Request register 0x0220.2: 0: Disabled 1: Enabled
			[12:13]	SYNC1 output driver/polarity: 00: Push-Pull active low 01: Open Drain (active low) 10: Push-Pull active high 11: Open Source (active high)
			[14]	SYNC1/LATCH1 configuration: 0: LATCH1 input 1: SYNC1 output
			[15]	SYNC1 mapped to AL Event Request register 0x0220.3: 0: Disabled 1: Enabled
0X2	Pulse Length of SYNC Signals: Initialization value for Pulse Length of SYNC Signals register	0X0982: x0983	[15:0]	Pulse length of SyncSignals (in Units of 10ns) 0: Acknowledge mode: SyncSignals will be cleared by reading SYNC[1:0] Status register
0X3	Extended PDI Configuration: Initialization value for extended PDI Configuration register	0x0152:0x0153 Digital I/O (0x140:04)	[0]	Direction of I/O[1:0]
			[1]	Direction of I/O[3:2]
			[2]	Direction of I/O[5:4]
			[3]	Direction of I/O[7:6]
			[4]	Direction of I/O[9:8]
			[5]	Direction of I/O[11:10]
			[6]	Direction of I/O[13:12]
			[7]	Direction of I/O[15:14]
			[8]	Direction of I/O[17:16]
			[9]	Direction of I/O[19:18]
			[10]	Direction of I/O[21:20]
			[11]	Direction of I/O[23:22]
			[12]	Direction of I/O[25:24]
			[13]	Direction of I/O[27:26]
			[14]	Direction of I/O[29:28]
			[15]	Direction of I/O[31:30]
	Extended PDI Configuration: Initialization value for extended PDI Configuration register	0x0152: 0x0153 SP I(0x140:0x5) SQI (0x140:0x83)	[0]	Direction of I/O[1:0]
			[1]	Direction of I/O[3:2]
			[2]	Direction of I/O[5:4]
			[3]	Direction of I/O[7:6]
			[4]	Direction of I/O[9:8]
			[5]	Direction of I/O[11:10]
			[6]	Direction of I/O[13:12]
			[7]	Direction of I/O[15:14]
			3:0	Length of byte(s) from Address to the first Write access Length = Dummy bytes + 2(Address bytes) – 1 Min Value :1
			7:4	Length of byte(s) from Address to the first Read access Length = Dummy bytes + 2(Address bytes) + 1(Length bytes) – 1 Min Value :4
		0x0152:	[0]	Read BUSY delay:

		0x0153 HBI (0x140: 0x8/0x9)		0: Normal read BUSY output 1: Delayed read BUSY output
			[1]	Perform internal write at: 0: End of write access 1: Beginning of write access
			[15:2]	Reserved, set EEPROM value 0
		0x0152: 0x0153 HBI (0x140: 0x90/ 0x97)	[0]	Read WAIT_ACK Removal Delay 0: Normal read WAIT_ACK output 1: Delayed read WAOT_ACK output
			[1]	Perform Internal Write 0: Following host write cycle (posted) 1: Following host read cycle (non-posted)
			[2]	HBI BE1/BE0 Polarity 0: BE1/BE0 pins active low 1: BE1/BE0 pins active high Note: Multiplexed Mode only
			[3]	IRQ output polarity: 0: active low 1: active high
			[4]	HBI BE1/BE0 Ignore 0: BE1/BE0 pins activated 1: BE1/BE0 pins ignored, value is 0
			[15:5]	RESERVED
0x4	Configured Station Alias: Initialization value for Configured Station Alias Address register	0x0012: 0x0013	[15:0]	Alias Address used for node addressing (FPxx commands). The use of this alias is activated by Register DL Control Bit 24 (0x0100.24/0x0103.0) NOTE: EEPROM value is only taken over at first EEPROM load after power-on or reset.
0x5	ASIC Configuration Register	0x0142: 0x0143	[4:0]	Digital input filter time: 0x00: off 0x01: 100us 0x02: 200us 0x03: 500us 0x04~0x1F: 1ms~28ms(one step plus 1ms)
			[5]	Reserved, set EEPROM value 0
			[6]	STATE_RUNLED Mode Select: 0: disabled 1: enabled
			[7]	Reserved, set EEPROM value 0
			[9:8]	When Watchdog expiration, Digital Output: 00: High Resistance 01: Keep last data 10: Zero 11: One
			[13:10]	Reserved, set EEPROM value 0
			[14]	ERRLED Enable: 0: disabled 1: enabled
			[15]	Reserved, set EEPROM value 0
0x6	Reserved		[15:0]	Reserved